world without plate tectonics, the sculpting of landforms on a planet with sandstorms undreamt of on Earth, glaciers and polar landforms, the escape of planetary atmospheres, and the capture of moons—to mention a more or less random sampling of scientific puzzles. If Mars once had abundant liquid water and a clement climate, what went wrong? How did an Earthlike world become so parched, frigid, and comparatively airless? Is there something here we should know about our own planet?

We humans have been this way before. The ancient explorers would have understood the call of Mars. But mere scientific exploration does not require a human presence. We can always send smart robots. They are far cheaper, they don't talk back, you can send them to much more dangerous locales, and, with some chance of mission failure always before us, no lives are put at risk.

"HAVE YOU SEEN ME?" the back of the milk carton read. "*Mars Observer*, 6' × 4.5' × 3', 2500 kg. Last heard from on 8/21/93, 627,000 km from Mars."

"*M. O.* call home" was the plaintive message on a banner hung outside the Jet Propulsion Laboratory's Mission Operations Facility in late August 1993. The failure of the United States' *Mars Observer* spacecraft just before it was to insert itself into orbit around Mars was a great disappointment. It was the first post-launch mission failure of an American lunar or planetary spacecraft in 26 years. Many scientists and engineers had devoted a decade of their professional lives to *M. O.* It was the first U.S. mission to Mars in 17 years—since *Viking*'s two orbiters and two landers in 1976. It was also the first real post—Cold War spacecraft: Russian scientists were on several of the investigator teams, and *Mars Observer* was to act as an essential radio relay link for landers from what was then scheduled to be the Russian *Mars '94* mission, as well as for a daring rover and balloon mission slated for *Mars '96*.

The scientific instruments aboard *Mars Observer* would have mapped the geochemistry of the planet and prepared the way for future missions, guiding landing site decisions. It might have cast a